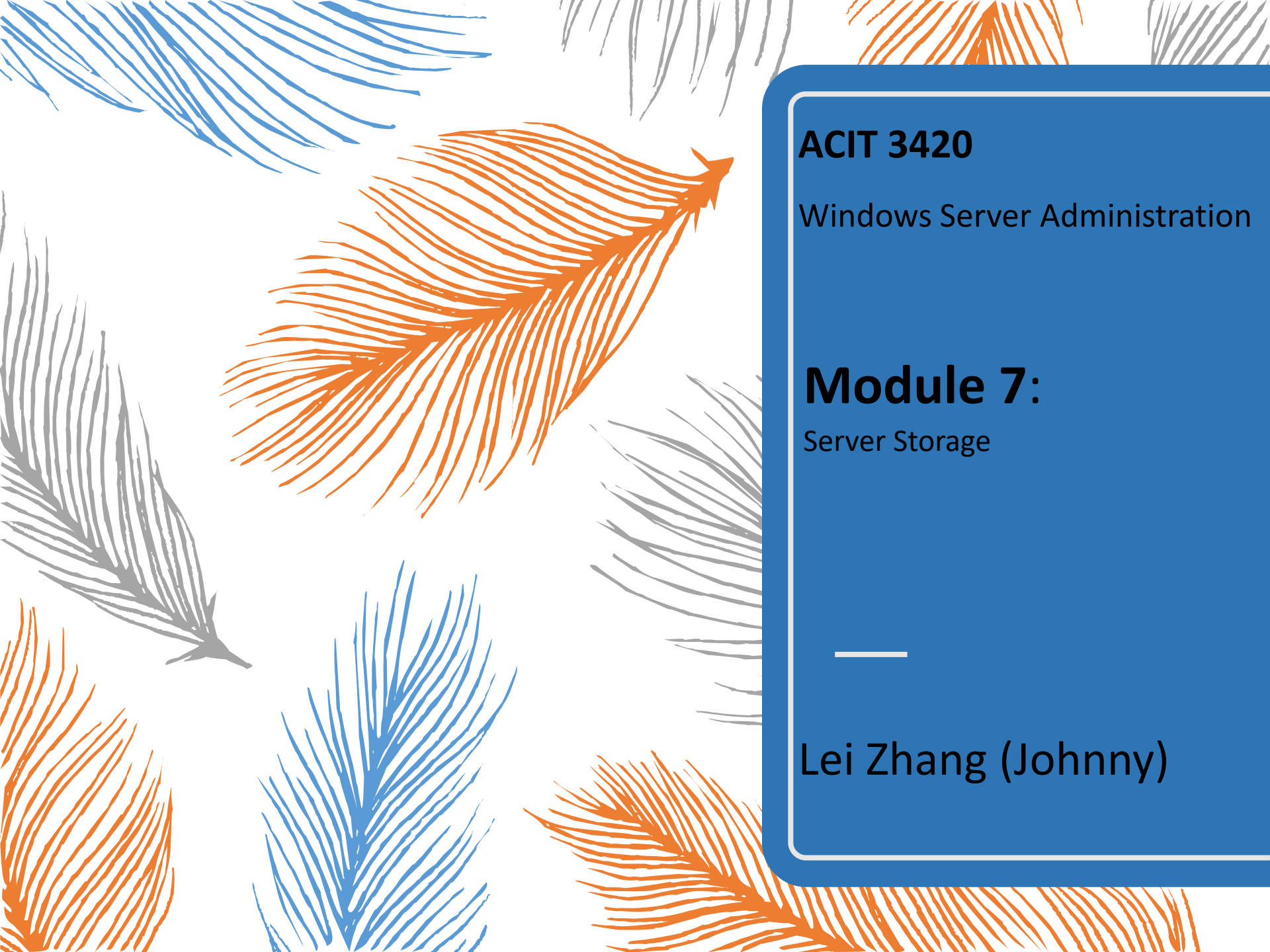


ACIT 3420

Windows Server Administration

Module 7:

Server Storage

Lei Zhang (Johnny)

Configuring Disks and Drive Types

- In Windows Server 2019/2022, a physical hard drive can be designated as a basic disk or a dynamic disk.
  - **Basic Disks**
    - ✓ *Default storage for Windows OS*
    - ✓ *Contain only simple volumes*
  - **Dynamic Disks**
    - ✓ *Can be modified without restarting the Windows system*
    - ✓ *Can contain simple, spanned, striped, and mirrored volumes*
- A ***partition*** is a defined storage space on a hard disk. To be usable, a hard disk needs to have at least one partition.
- A ***volume*** is a partition that has been formatted into a file system (Disk volume requirements include a system volume and a boot volume).

Configuring Disks and Drive Types

You can configure dynamic disk volumes as:

- ***Simple volume***: Uses free space available on a single disk.
- ***Spanned volume***: Extends a simple volume across multiple disks, up to a maximum of 32.
- ***Mirrored volume***: Duplicates data from one disk to a second disk for redundancy and fault tolerance; if one disk fails, data can be accessed from the second disk. You cannot span a mirrored volume; a mirrored volume must reside on a single disk. Mirroring is also referred to as RAID-1.
- ***Striped volume***: Stores data across two or more physical disks. Data on a striped volume is written evenly to each of the physical disks in the volume. You cannot mirror or span a striped volume. Striping is often referred to as RAID-0.

Configuring local storage: Selecting a Partition Table Format

MBR: Master Boot Record

- Standard partition table format since early 1980s
- Supports a maximum of four primary partitions per drive or up to three primary partitions with one extended partition.
- Can partition a disk up to 2 TB
- The MBR is a single point of failure. If it becomes corrupted or damaged, it could prevent the computer from starting.

GPT

- GPT is the successor of the MBR partition table format
- Supports a maximum of 128 partitions per drive
- Can partition a disk up to 18 exabytes

- **Use MBR for disks smaller than 2 TB**
- **Use GPT for disks larger than 2 TB**

Selecting a File System

When selecting a file system, consider the differences between FAT, NTFS, and ReFS

- FAT provides:
  - Basic file system
  - Partition size limitations
  - FAT32 to enable larger disks
  - exFAT developed for flash drives
- NTFS provides:
  - Metadata
  - Auditing and journaling
  - Security (ACLs and encryption)
- ReFS provides:
  - Backward compatibility support for NTFS
  - Enhanced data verification and error correction
  - Support for larger files, directories, and volumes

Implementing ReFS (Advantages)

- Metadata integrity with checksums
- Integrity streams with user data integrity
- Allocation on write transactional model
- Large volume, file, and directory sizes (2^78 bytes with 16 KB cluster size)
- Storage pooling and virtualization
- Data striping for performance and redundancy
- Disk scrubbing for protection against latent disk errors
- Resiliency to corruptions with recovery
- Shared storage pools across machines

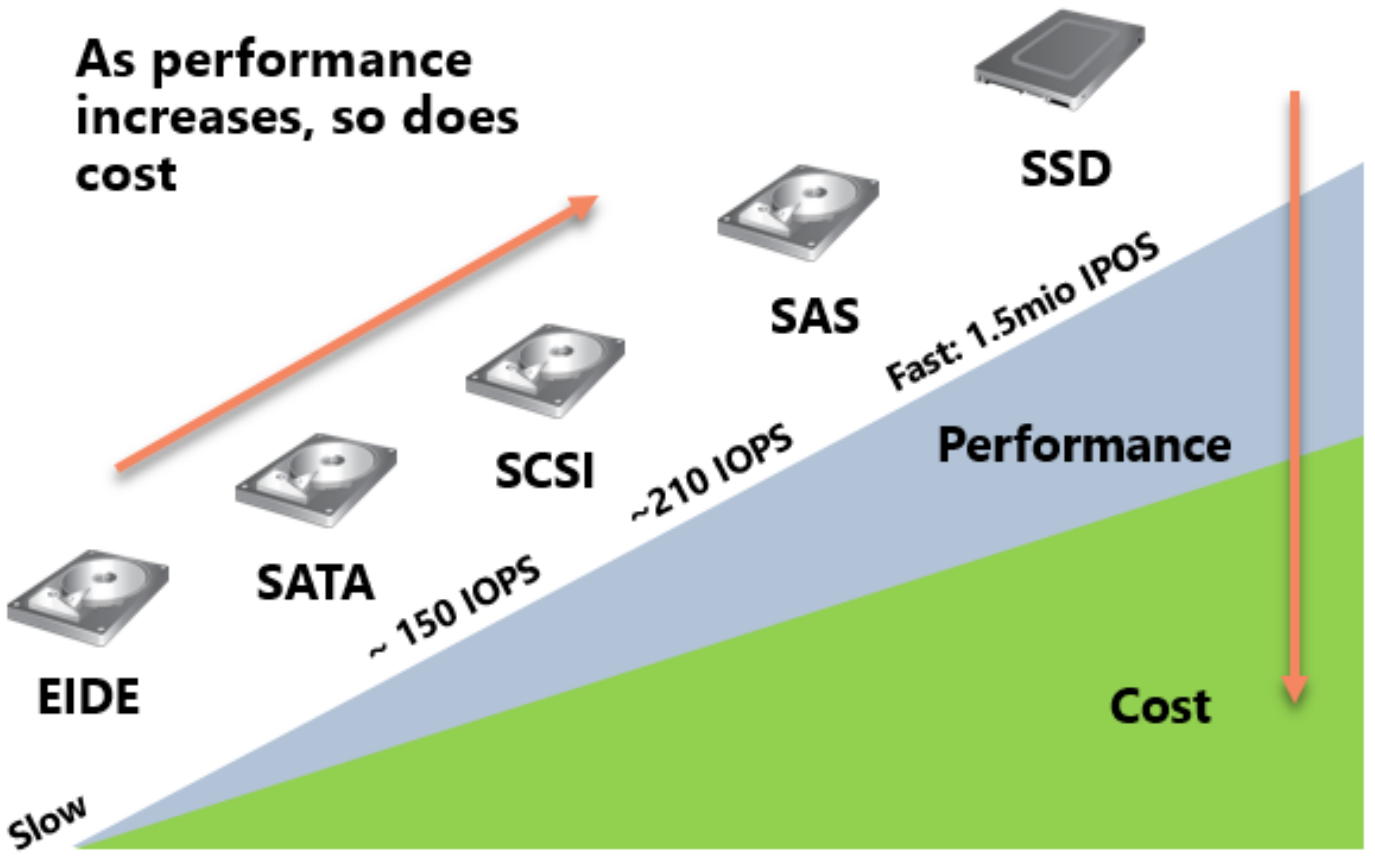
```
PS C:\windows\system32> fsutil fsinfo volumeinfo c:
Volume Name : WinOS
Volume Serial Number : 0x42169172
Max Component Length : 255
File System Name : NTFS
Is ReadWrite
Supports Case-sensitive filenames
Preserves Case of filenames
Supports Unicode in filenames
Preserves & Enforces ACL's
Supports file-based Compression
Supports Disk Quotas
Supports Sparse files
Supports Reparse Points
Returns Handle Close Result Information
Supports POSIX-style Unlink and Rename
Supports Object Identifiers
Supports Encrypted File System
Supports Named Streams
Supports Transactions
Supports Hard Links
Supports Extended Attributes
Supports Open By FileID
Supports USN Journal
```

```
PS C:\windows\system32> fsutil fsinfo sectorinfo c:
LogicalBytesPerSector :                                 512
PhysicalBytesPerSectorForAtomicity :                    512
PhysicalBytesPerSectorForPerformance :                  512
FileSystemEffectivePhysicalBytesPerSectorForAtomicity : 512
Device Alignment :                                      Aligned (0x000)
Partition alignment on device :                         Aligned (0x000)
No Seek Penalty
Trim Supported
Not DAX capable
```

```
PS C:\Users\administrator.ADATUM> Format-Volume -DriveLetter E -FileSystem refs

DriveLetter FileSystemLabel FileSystem DriveType HealthStatus OperationalStatus SizeRemaining    Size
----------- --------------- ---------- --------- ------------ ----------------- -------------    ----
E                           ReFS       Fixed     Healthy      OK                      9.22 GB 9.94 GB
```

Selecting a Disk Type

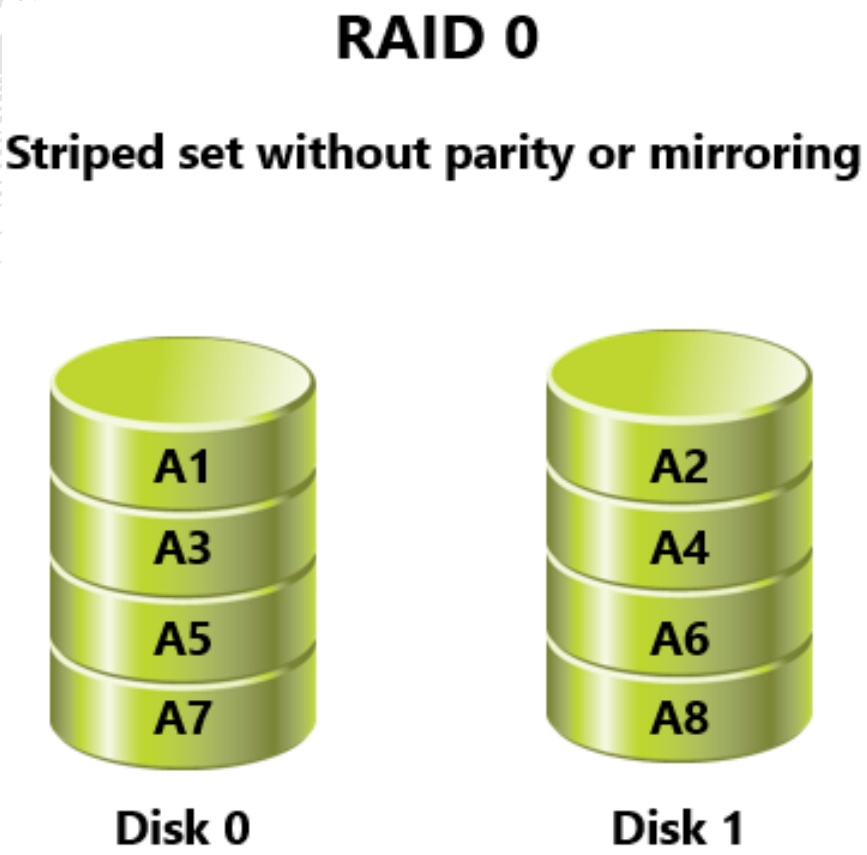


What is RAID

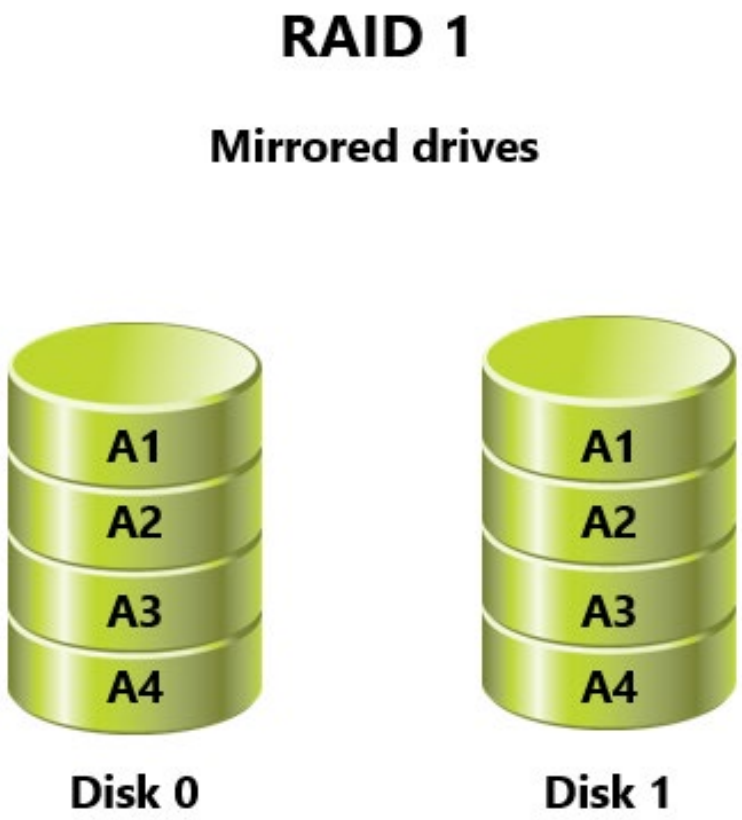
RAID:

- Combines multiple disks into a single logical unit to provide fault tolerance and performance
- Provides fault tolerance by using:
  - ✓Disk mirroring
  - ✓Parity information
- Can provide performance benefits by spreading disk I/O across multiple disks
- Can be configured using several different levels
- Should not replace server backups

RAID Levels: RAID 0

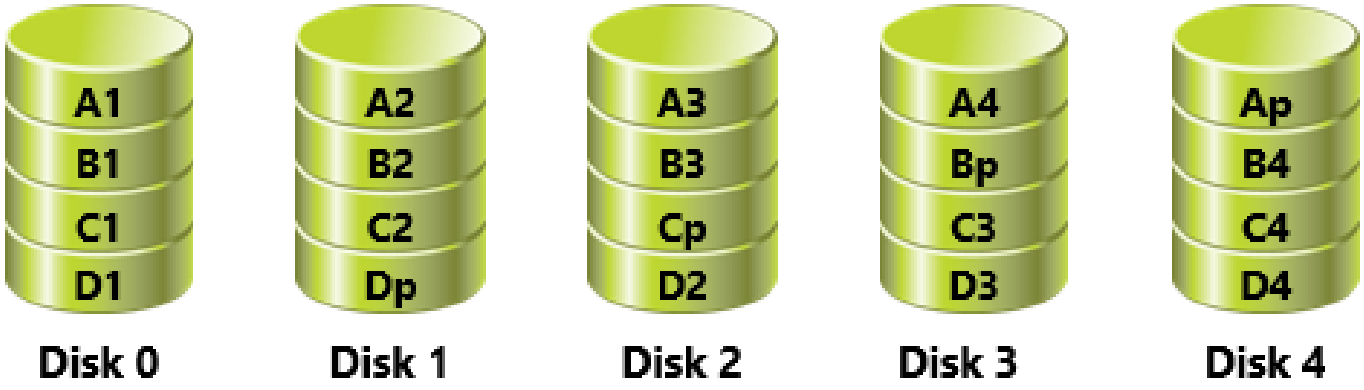


RAID Levels: RAID 1

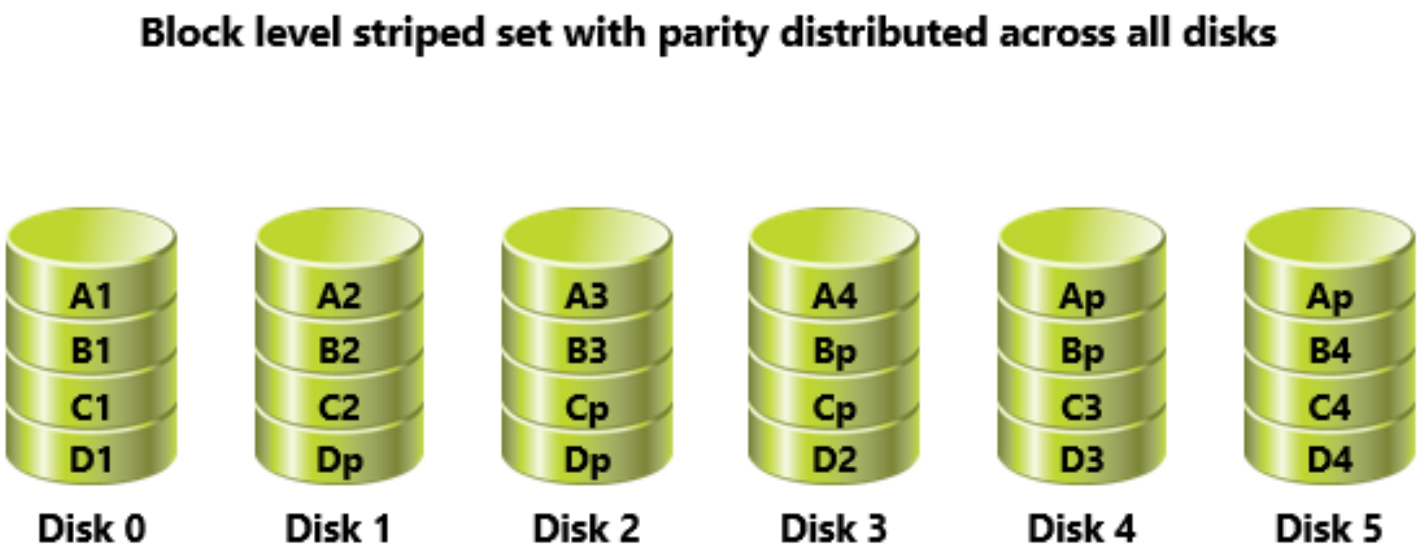


RAID Levels: RAID 5

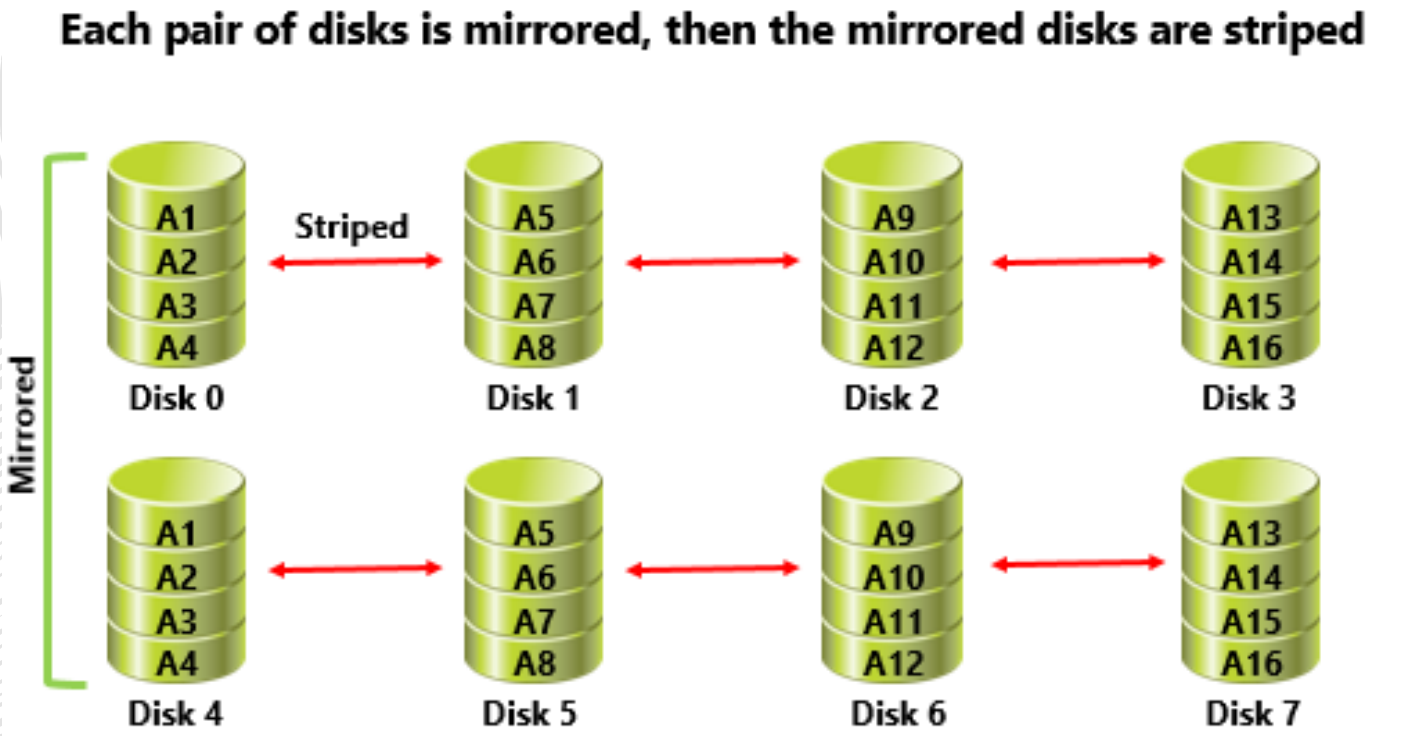
Block level striped set with parity distributed across all disks



RAID Levels: RAID 6



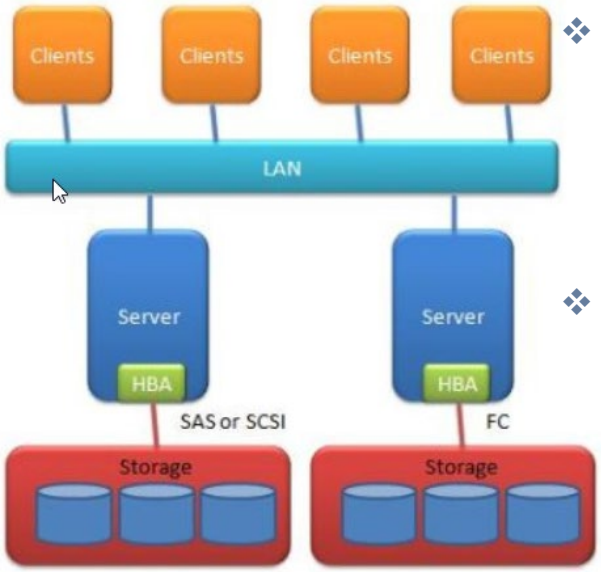
RAID Levels: RAID 1+0



Overview of Enterprise Storage Solutions

- When planning storage for Windows Server 2019/2022, you need to determine how the servers will access the disks and multiple options exist:
  - Direct Attached Storage(DAS)
  - Network Attached Storage(NAS)
  - Storage Area Network(SAN)
- These options differ in type as well as access technologies and usage scenarios
- The right choice is critical to the overall success of the storage solution.

Direct Attached Storage



- DAS is a technology that is available in nearly every server
  - Can include
    - *Physical disks located inside the server*
    - *Connections to an external array*
    - *Connections to USB disks or alternative connectors.*
  - Advantages (Easiest storage system to deploy and Maintain. Example? ERP)
    - *Easy to configure (Disk Management)*
    - *In expensive (Based on disk type and host bus adapter (HBA)) (No Network devices, hubs, switches or routers)*
  - Disadvantages
    - *Isolated because the disks are attached to a single server*
    - *Less flexibility*
    - *Scalability limits*

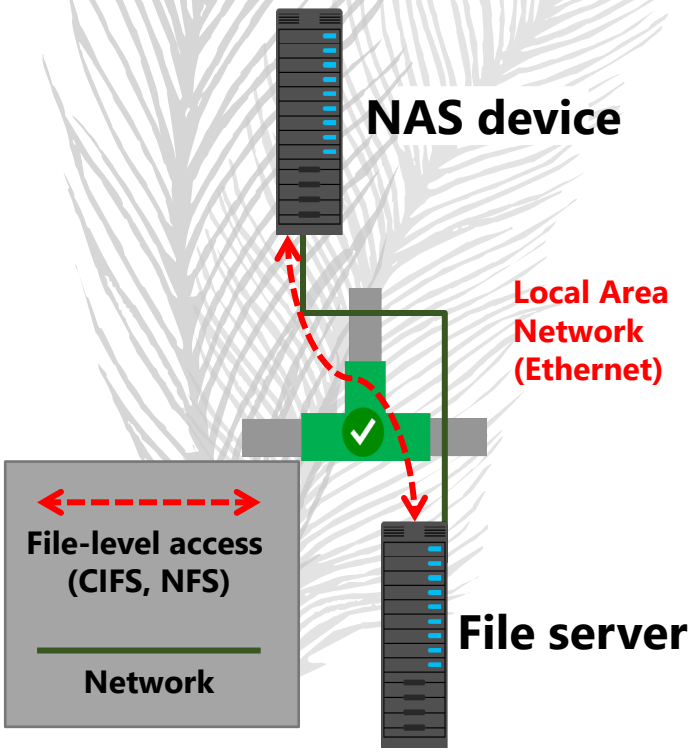
NAS


WD
My Cloud
EX4100
Welcome to
WD My Cloud




HPE StoreEasy Storage
Focus on your business, not your file storage
StoreEasy 1860
Hewlett Packard
Enterprise

Network Attached Storage

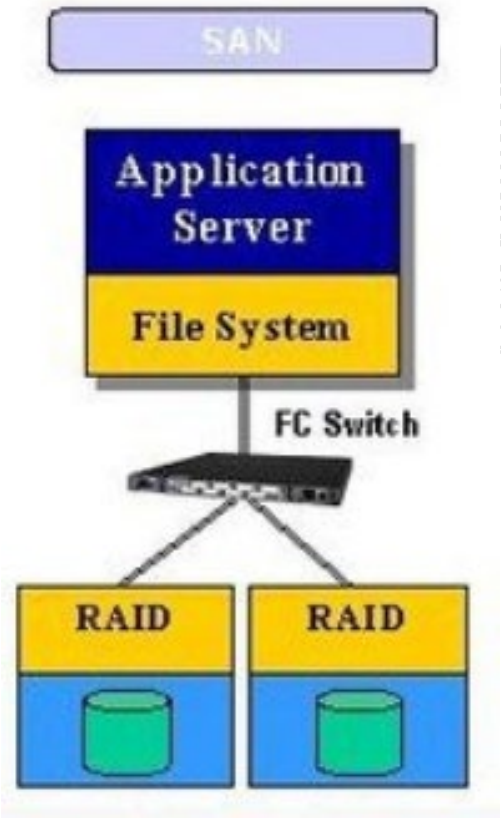


- NAS is a file-level data storage device that is connected to the server over a computer network to provide shared drives or folders, usually using Server Message Block (SMB) or Network File System (NFS).
  - Storage locations are accessed through file level access to shares
  - Use protocols like (Common Internet File System)CIFS, (Server Message Block) SMB, and (Network File System) NFS to access files.
  - Devices are often in headless configuration requiring network communication in order to configure the device

Advantages vs. Disadvantages

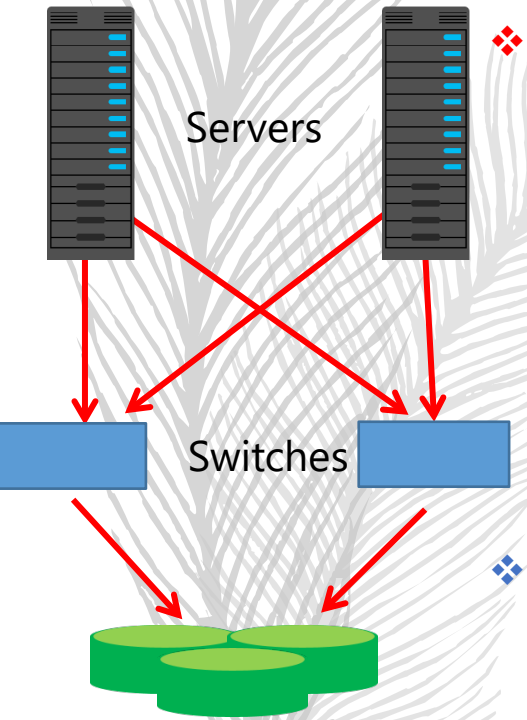
- Advantages
  - Relatively inexpensive centralized storage
  - Easy to configure
- Disadvantages
  - Slower access times
  - Not appropriate in many enterprise scenarios (Sql, Exchange)

Storage Area Network (SAN)



- A SAN is an enterprise centralized storage solution that provides **high availability** and maximum **flexibility**.
- SAN is a storage architecture that allows systems to attach to the storage in the SAN and presents the drives to the server just as if the drives were locally attached.
  - Include various components
  - Provides block level access to centralized storage

SAN: Advantages vs. Disadvantages



- **Advantages**
  - Storage can be accessed by multiple servers simultaneously with more robust, faster processing.
  - Easily expandable
  - Centralized storage with high levels of redundancy
  - High level of redundancy(Multiple network devices, such as switches. Redundant components such as Uninterruptible power supply, hard drives)
- **Disadvantages**
  - Most expensive solution (A entry-level SAN=Full loaded server with a DAS or NAS device)
  - Requires specialized administrative skills (Logical Unite Number (LUN) setup, Each storage vendors provide different tools and features)

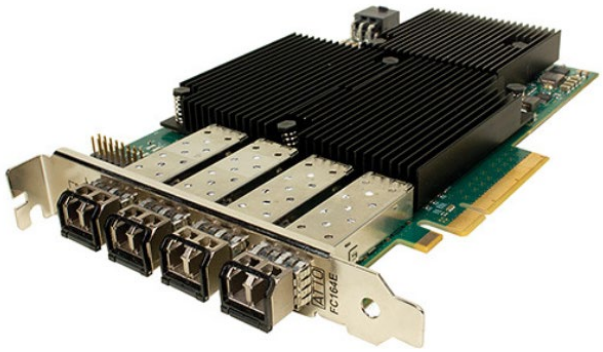
Protocol to Configure SAN: Fibre Channel

- Fibre Channel is high performance network technology primarily used within SAN solutions
- **Node**: Any device connected to the SAN.
- **WWN**(**World Wide Name**): Unique identifier used to identify storage devices
- **Fabric**: Encompasses all hardware that connects servers and workstations to storage devices through the use of fibre channel switching technology

FC Components

- **Components**
  - SAN (Storage backend; It serves as a Fibre Channel target which is the component that listens for requests from computers.)
  - Physical or virtual system containing an HBA card (FC or iSCSI)
  - Fiber Channel Switch
- Fibre Channel over Ethernet (**FCoE**) (offering outstanding performance: in expensive, preexisting, Ethernet infrastructure)
  - Provides the benefits of Fibre Channel in a more cost-effective manner
  - It is easier to manage a single network topology
  - Specialized training is not required.

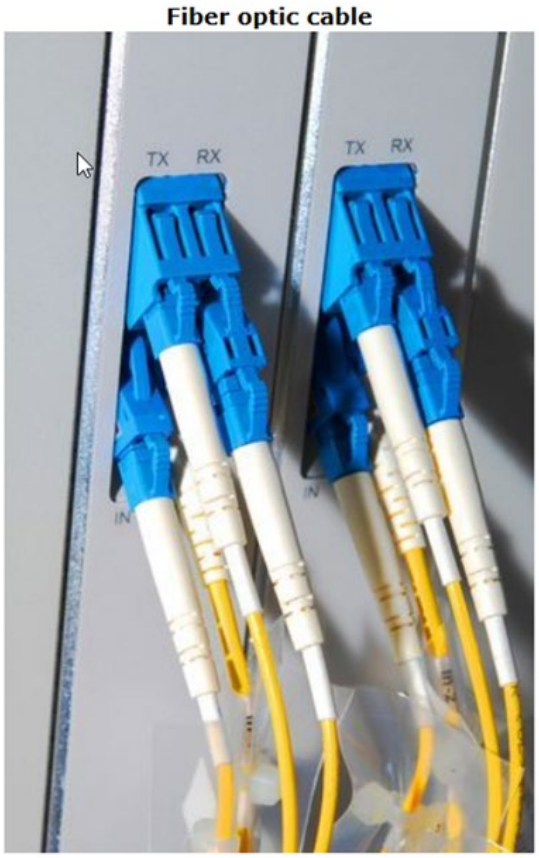






Fiber Optic Cable

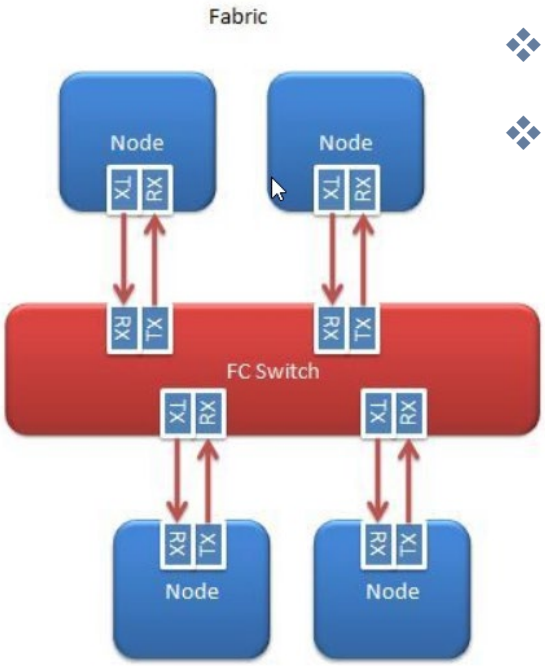
Fibre channel transmits data serially, this means bit by bit. That's why the cabling only consists of a transmit port and a receive port.

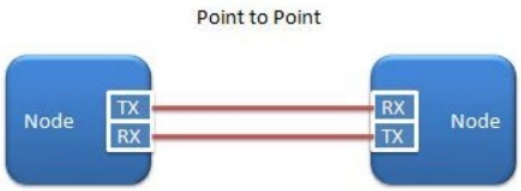


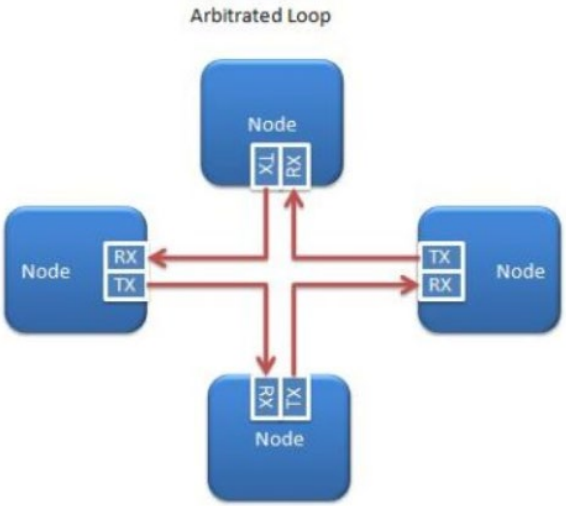
Fibre Channel Layouts (O)

You can arrange a Fibre Channel network in one of three layouts

- Arbitrated Loop
- Point-to-point
- Switched Fabric







Fibre Channel Considerations

There are several important factors used to determine whether Fibre Channel is right for your environment

- Infrastructure requirements
  - *Fabric or network switches*
  - *HBAs*
  - *Additional cabling*
  - *Storage controllers*
- Bandwidth requirements (32Gbps)
- Reliability (TCP/IP: TTL lost packages) and security (Distributed denial of service (DDos))
- Costs
  - *More expensive than other solutions*
  - *Large initial investment*
  - *Training costs are often high*

Overview: Comparing Storage Solutions

- DAS
  - Least complex
  - Low setup costs
- NAS
  - Best solution in specific situations
  - Often used to compliment DAS and SAN solution
- SAN
  - Highest levels of performance
  - Provides the most features
- Windows Server 2019/2022 provides storage features that are expanding to improve capabilities when using DAS.

Comparing Storage Technology

- You can arrange data on a disk in two ways: by blocks or by files
- Block level storage
  - High performing
    - Most often associated with SANs
    - Presents Logical Unit Numbers (LUNs) to the Server
    - Not the most cost effective
- File level storage
  - Delivered via NAS, an external storage or file server
  - Uses common file level protocols (CIFS, SMB, NFS)
  - Uses block level storage on the backend

Scenarios for using DAS, NAS or SAN

- High performance storage for transactional databases
- User home folders
- Storage of virtual machines
- Branch office shared folders
- Tiered storage for applications
- Microsoft exchange database and log storage

iSCSI SAN

- iSCSI (Internet) is a protocol that supports access to SCSI based storage over standard TCP/IP network (Note: Local network but it runs over Internet Protocol, TCP/IP)
  - Typically relies on standard Ethernet network architecture
  - Could use specialized hardware such as HBA or network switches
  - Uses TCP/IP and TCP port 3260 to communicate
- Components
  - IP network
  - iSCSI targets: servers that run on the storage device
  - iSCSI initiators: software or host adapter that provides access to targets
  - IQN (iSCSI Qualified Name): Globally unique identifier used to address initiators and targets on an iSCSI network

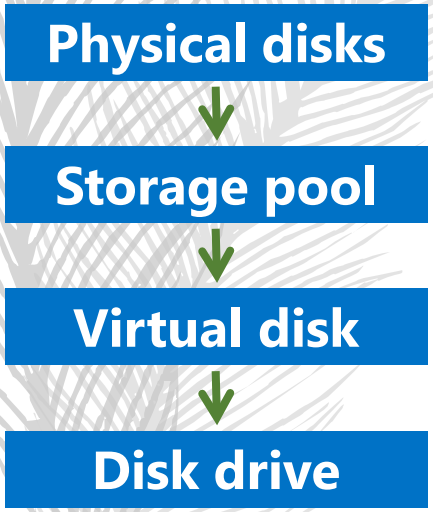
iSCSI Target Server

- iSCSI Target Server
  - Available as a role service in Windows Server 2019/2022
  - Provides different types of functionality
    - *Network or diskless boot*
    - *Server application storage*
    - *Non-Microsoft platforms*
    - *Lab environments*
  - Features
    - *Authentication (Challenge handshake authentication protocol)*
    - *Query initiator computer for ID*
    - *Virtual hard disks*
    - *Scalability and manageability*

iSCSI Initiator

- iSCSI initiator
  - Runs as an operating system service
  - Installed by default in Windows Server 2008 and Vista and later operating systems
  - Only requires the service to be started before configuring the initiator to connect to iSCSI target.

Configuring Storage Pools and Storage Spaces



- *Storage Spaces (Virtual Disks)* is a storage virtualization feature built into Windows Server 2019/2022 and Windows 10.
- The Storage Spaces feature consists of two components:
  - ✓ Storage Pools are a collection of any type and size of physical disks (JBOD) aggregated into a single logical disk, allowing you to manage the multiple physical disks as a single disk.
  - ✓ These storage pools can then be segmented into multiple storage spaces, which are then formatted with a file system and can be used just like any other regular disk on your computer.
- New disks (internal/external) can be added to the storage pool as space requirements increase over time.

VHDX vs. VHD

Creating Storage Pools

- One physical disk is required to create a storage pool
- At least two physical disks are required to create a resilient, mirrored virtual disk.
- At least three physical disks are required to create a virtual disk with resiliency through parity.
- At least five physical disks are required for three-way mirroring
- At least seven physical disk are required for dual parity.
- Disk must be blank and unformatted, which means no volumes can exist on the disks.
- Disks can be attached using a variety of bus interfaces including SAS, SATA, SCSI, NVMe, and USB.

Creating and Configuring Storage Spaces

- When creating storage spaces, there are four resiliency types to select from (Storage Layout). Only three of them provide real fault tolerance, as shown in the following list:
- ***Simple (no resiliency)***: Writes one copy of your data but doesn't protect against drive failures; requires at least one drive (RAID 0)
- ***Two-way mirror***: Writes two copies of your data to protect against a single drive failure; requires at least two drives (RAID 1)
- ***Three-way mirror***: Writes three copies of your data to protect against two simultaneous drive failures; requires at least five drives
- ***Parity***: Writes data with parity information to protect against single drive failures; requires at least three drives (RAID 5)

Considerations for Implementing iSCSI

- Several factors exist which should be examined prior to the implementation of iSCSI
  - Network speed and performance
  - High availability
  - Security
  - Infrastructure staff
  - Application teams
  - Vendor information
- Consider alternative solutions

iSNS

- The ***Internet Storage Name Service (iSNS)*** protocol is used to discover, manage, and configure iSCSI devices automatically on a TCP/IP network.
- iSNS is a Windows feature containing three primary components.
  - iSNS Server (respond to iSNS protocol queries and requests, authentication)
  - iSNS client
  - iSNS database (an information repository, contains information about iSNS client attributes)
- Provides numerous advantages
  - Reduces administrative overhead
  - Compatible with multiple storage technologies
  - Can make an IP network function like a SAN
  - Can integrate IP and Fibre Channel networks

iSNS Functions and Installation

- The four primary functions of iSNS are
  - A name service
  - A discovery domain and login control service
  - A state change notification service
  - Mapping information to an iSNS database
- Installation occurs using Server Manager or Windows PowerShell

MPIO

- A single connection to an iSCSI storage device makes the storage available, but it does not make the storage connections highly available.
- If the network connection or a switch fails, the server connecting to the iSCSI storage will lose access to its storage.
- Because many servers require high availability, you need to use high-availability technologies, such as multiple connected session (MCS) and multi-path I/O (MPIO).
- Built into Windows Server since Windows Server 2008
- Storage vendors offer their own MPIO software
- Used primarily
  - Create a highly available storage infrastructure
  - Increase performance to achieve high performance requirements.

Configuring Sharing in Windows Server 2019/2022

- File sharing is a core service on any network and provided in Windows Server 2019/2022 by default
- Each version of Windows Server will increase upon the capabilities of the previous versions
  - SMB version change
  - NFS updates
  - FSRM capabilities
  - Storage of VHDs on an SMB share
  - Scaled out File server cluster

Server Message Block

- SMB is the file sharing protocol that Windows clients and servers will utilize and comes in different versions
- Versions
  - SMB 1.X : Currently no dependencies on this antiquated version and it can be disabled in most situations
  - SMB 2.X: supported in Windows Vista and Windows 7
  - SMB 3.0 : supported in Windows 8 and Server 2012
  - SMB 3.0.2: supported in Windows 8.1 and Server 2012R2
  - SMB 3.1.1: supported in Windows 10 and Windows Server 2019/2022

Configuring SMB Shares

- Folders must be shared in order to be accessible over the network
- Windows Server 2019/2022 provides various tools to create shared folder
  - Server Manager
  - Windows PowerShell
  - File explore
- SMB share profiles
  - Quick
  - Advanced
  - Applications

Configuring SMB shares

- Windows PowerShell cmdlets for SMB share management:
  - **New-SmbShare**
  - **Set-SmbShare:** is used to modify the properties of the Server Message Block (SMB) share
  - **Remove-SmbShare**
  - **Get-SmbShare:** is used to retrieve retrieves objects that represent the Server Message Block (SMB) shares being displayed by the computer.
  - **Get-SmbSession:** is used to identify users that are connected to SMB shares
  - **Get-SmbOpenFile:** is used to identify open files if the users have open files.
  - **Set-SmbBandwidthLimit:** is used to define a maximum throughput level that is allocated to SMB traffic on a server for different categories. (useful for Hyper-V hosts.)

What is NFS

- NFS is a file system based on open standards
- Current version is 4.1
- Windows NFS components include:
  - Client for NFS
  - Server for NFS
- Support for Kerberos v5 authentication
- The primary uses for NFS are:
  - Storage for VMware virtual machines
  - Sharing data across multiple operating systems
  - Sharing data across different IT infrastructures after a company merger

Configuring NFS shares

- Install the Server for NFS server role
- Two options for NFS share profile:
  - NFS Share – Quick
  - NFS Share – Advanced
- Authentication options:
  - Kerberos v5 authentication
  - No server authentication
- Share permissions define allowed and denied hosts
- Follow best practices